

PROJECT REPORT

UPI Transaction Fraud Detection Analytics – India

Power BI Dashboard Analysis Report

1. Project Title

UPI Transaction Fraud Detection Analytics – India

Domain: Banking & Financial Services (BFSI)

Technology Stack:

- Power BI
 - SQL
 - Python
 - Excel / Power Query
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2. Executive Summary

The rapid adoption of the Unified Payments Interface (UPI) has transformed India's digital payment ecosystem by enabling fast, secure, and cashless transactions. However, the increasing transaction volume has also elevated the risk of fraudulent activities, making fraud detection and risk monitoring critical for financial institutions.

This project focuses on analyzing **250,000 UPI transactions** to identify transaction patterns, fraud indicators, geographical trends, banking behavior, and operational risks. The interactive Power BI dashboard provides a centralized monitoring system that enables stakeholders to evaluate transaction performance while simultaneously monitoring fraud exposure.

The dashboard combines business performance metrics with fraud analytics, allowing management teams to make informed decisions regarding transaction security, operational efficiency, and fraud prevention strategies.

The insights generated from this project support financial institutions in improving fraud detection capabilities, reducing financial losses, strengthening monitoring frameworks, and enhancing customer trust.

3. Business Problem Statement

India processes millions of UPI transactions every day across multiple banks, devices, merchant categories, and geographic regions. Even a small percentage of fraudulent transactions can lead to substantial financial losses and reputational damage.

The objective of this project is to analyze UPI transaction data to identify:

- Fraudulent transaction patterns
- High-risk states
- High-risk banking networks
- Transaction behavior across different time periods
- Operational indicators that can improve fraud detection

These objectives align with the business goals described in the project statement.

4. Business Objectives

The primary objectives of this project are:

- Monitor overall UPI transaction performance.
 - Measure fraud occurrence across the transaction ecosystem.
 - Identify high-risk geographic regions.
 - Analyze transaction behavior across banks.
 - Study transaction value distribution over different time periods.
 - Evaluate category-wise transaction contribution.
 - Support proactive fraud monitoring.
 - Enable data-driven financial decision making.
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5. Dataset Overview

The dataset consists of **250,000 synthetic UPI transaction records** containing transaction details such as:

- Transaction Amount
- Transaction Status
- Fraud Flag
- Sender Bank
- Receiver Bank
- Sender State
- Merchant Category
- Device Type
- Network Type
- Transaction Timestamp
- Time Bucket
- Day of Week
- Age Groups

The dataset was designed to simulate real-world UPI transaction scenarios for fraud analytics.

6. Dashboard Overview

The dashboard provides a consolidated view of transaction performance and fraud monitoring.

It contains:

KPI Cards

- Total Transaction Value
- Total Transactions
- Average Transaction Amount
- Total Fraud Transactions
- Failed Transactions
- Fraud Rate

Trend Analysis

- Monthly Transaction Trend

Distribution Analysis

- Transaction Value by Time Bucket
- Transaction Value by State
- Transaction Contribution by Category
- Transaction Volume by Sender Bank

Interactive Filters

- Transaction Status
- Quarter Selection

7. Dashboard Analysis

7.1 Overall Business Performance

The dashboard indicates strong transaction activity across the platform.

Key Findings

KPI	Value
Total Transaction Value	₹327.94 Million
Total Transactions	250K
Average Transaction Amount	₹1.31K
Fraud Transactions	480
Failed Transactions	12K
Fraud Rate	0.19%

Business Interpretation

- The UPI platform processed over **250,000 transactions**.
- Total payment volume exceeded **₹327 Million**, indicating significant platform usage.
- The fraud rate is only **0.19%**, demonstrating effective fraud prevention while still requiring continuous monitoring.
- Approximately **12,000 transactions failed**, indicating opportunities for operational improvements.

7.2 Monthly Transaction Trend

The monthly trend illustrates transaction volume fluctuations throughout the year.

Observations

- Monthly transaction values remain relatively stable.
- Peaks are observed around **May, July, and August**.
- Minor declines occur during **February and November**.
- Overall transaction volume remains consistent.

Business Insight

Stable transaction behavior indicates consistent customer engagement and healthy platform utilization.

7.3 Transaction Value Distribution by Time Bucket

The dashboard categorizes transaction values into:

- Morning
- Afternoon
- Evening
- Night

Findings

- **Evening** contributes the highest transaction value.
- Afternoon records the second-highest transaction volume.
- Morning transactions remain moderate.

- Night transactions contribute the lowest share.

Business Insight

Evening hours represent peak customer activity and therefore require stronger fraud monitoring due to higher transaction concentration.

7.4 State-wise Transaction Value

The dashboard compares transaction values across Indian states.

Top Performing States

- Maharashtra
- Uttar Pradesh
- Karnataka
- Tamil Nadu
- Delhi

Observations

Maharashtra contributes the largest transaction value, indicating high customer adoption and economic activity.

Business Insight

States with larger transaction volumes should receive enhanced fraud monitoring since greater transaction activity can increase fraud exposure.

7.5 Transaction Category Analysis

The dashboard evaluates transaction contributions across merchant categories.

Highest Categories

- Shopping
- Grocery
- Utilities

Lowest Categories

- Healthcare
- Entertainment
- Transport

Business Insight

Shopping contributes the highest transaction volume, making it an important category for fraud monitoring and customer behavior analysis.

7.6 Sender Bank Analysis

Transaction volumes are distributed across several banks.

Top Banks

- SBI
- HDFC
- ICICI
- IndusInd
- Yes Bank

Observations

SBI records the highest transaction volume among all participating banks.

Business Insight

Banks processing larger transaction volumes should implement more advanced fraud monitoring systems due to increased exposure.

7.7 Fraud Monitoring

The dashboard provides dedicated fraud KPIs.

Findings

- Fraud Transactions: **480**
- Fraud Rate: **0.19%**

Business Interpretation

Although fraud represents a small proportion of total transactions, the financial impact can still be significant because fraudulent transactions often involve higher-value payments and reputational risk.

8. Business Insights

The dashboard reveals several meaningful business insights:

Operational Insights

- Transaction activity remains stable throughout the year.
- Shopping dominates customer spending.
- Evening hours experience maximum payment activity.
- Maharashtra contributes the highest transaction value.
- SBI processes the largest transaction volume.

Risk Insights

- Fraud occurrence is relatively low.
 - High transaction-value regions require continuous monitoring.
 - Peak transaction periods demand stronger fraud surveillance.
 - Large banks require enhanced fraud detection mechanisms.
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9. Recommendations

Fraud Prevention

- Deploy real-time fraud detection models.
 - Monitor unusual transaction spikes.
 - Flag abnormal transaction amounts.
 - Detect repeated failed transactions.
 - Monitor suspicious sender-receiver relationships.
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Regional Risk Management

- Increase fraud surveillance in high-volume states.
 - Build region-specific fraud risk models.
 - Monitor interstate transaction anomalies.
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Bank-Level Improvements

- Strengthen fraud controls for high-volume banks.
 - Share fraud intelligence across banking partners.
 - Monitor bank-to-bank transaction patterns.
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Operational Improvements

- Reduce failed transaction rates through network optimization.
 - Improve payment gateway reliability.
 - Enhance customer authentication mechanisms.
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Customer Awareness

- Educate users about phishing attacks.
 - Encourage secure PIN practices.
 - Promote biometric authentication.
 - Notify users of suspicious transactions instantly.
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10. Suggested Future Enhancements

The dashboard can be further improved by incorporating:

Advanced Fraud Analytics

- Fraud Heat Maps
- Fraud Trend Forecasting
- Fraud Probability Score
- High-Risk Customer Identification
- Real-Time Fraud Alerts

AI & Machine Learning

- Anomaly Detection Models
- Predictive Fraud Scoring
- Behavioral Analytics
- Risk Classification Models
- Auto Fraud Alert System

Additional Dashboard Features

- Device-wise Fraud Analysis
 - Network-wise Fraud Distribution
 - Age Group Fraud Analysis
 - Weekday vs Weekend Fraud Trends
 - Transaction Type Fraud Analysis
 - Bank-to-Bank Fraud Matrix
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11. Business Impact

This dashboard enables financial institutions to:

- Monitor overall UPI ecosystem performance.
 - Identify fraudulent activities at an early stage.
 - Improve operational efficiency.
 - Reduce financial losses.
 - Strengthen fraud monitoring frameworks.
 - Support data-driven decision-making.
 - Enhance customer trust and payment security.
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12. Project Conclusion

The **UPI Transaction Fraud Detection Analytics Dashboard** successfully integrates business performance monitoring with fraud risk analytics into a single interactive platform. By analyzing **250,000 UPI**

transactions, the solution delivers actionable insights into transaction trends, state-wise activity, merchant categories, banking performance, and fraud exposure.

The dashboard demonstrates that while the overall fraud rate is relatively low (0.19%), continuous monitoring remains essential due to the scale of digital payment activity. It highlights high-volume states, peak transaction periods, and major banking contributors, enabling financial institutions to prioritize fraud prevention efforts where operational risk is greatest.

Overall, this project showcases how Business Intelligence (Power BI), SQL, and data analytics can support real-time fraud monitoring, enhance operational decision-making, strengthen risk management frameworks, and improve the security and reliability of India's rapidly growing digital payment ecosystem.